AN ELEMENTARY DIVISOR DOMAIN IS A BEZOUT DOMAIN

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot, namely sources/stacks-chow/preamble.tex lines 14-18 (Xy-pic setup) and 154-217 (theorem declarations and mathematical macros). Nothing was added, removed or redefined, no package was added, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Allowed standard prerequisites (editorial).

- Definition of an elementary divisor domain as one where every matrix can be diagonalised by invertible row and column operations. definition environment preceding the statement, source lines 37780-37812. Supplies the operation that produces the invertible u and V and the single entry f.
- The ideal generated by the entries of a column of an invertible matrix over a domain is the unit ideal. used at proof lines 37827-37829 to conclude (g_11, g_21) = R. Is the step that produces the Bezout relation, since it makes f a combination of a and b while a and b are combinations of f.
- A domain is Bezout if and only if every finitely generated ideal is principal. used in the omitted induction at proof line 37830. Reduces the statement to the two-element case that the author proves.

Source reference. Original lemma 0ASS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13590–13590. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Lemma 0.2. *An elementary divisor domain is Bézout.*

Proof. Let $a, b \in R$ be nonzero. Consider the 1×2 matrix $A = (a\ b)$. Then we see that $u(a\ b)V = (f\ 0)$ with $u \in R$ invertible and $V = (g_{ij})$ an invertible 2×2 matrix. Then $f = uag_{11} + ubg_{21}$ and $(g_{11}, g_{21}) = R$. It follows that $(a, b) = (f)$. An induction argument (omitted) then shows any finitely generated ideal in R is generated by one element. □

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.